Safety-Shielded Candidate-Graph Policy Improvement for Dynamics-Aware Orbital Inspection

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***Abstract*— Autonomous inspection of large orbital structures couples surface observability, long-horizon view selection, and constrained relative motion. We represent every decision by a safe observable orbital arc (SOOA): a finite-duration Hill–Clohessy–Wiltshire transfer-to-dwell action with camera-valid target set, control effort, clearance, terminal accuracy, and passive-drift evidence. The resulting candidate graph supports critic-guided lookahead, one-step rollout, audited local search, and an incumbent safeguard, while an HCW shield removes inadmissible transfers. On a public NASA ISS mesh with 180 weighted surface samples and 357 candidate views, the complete planner reaches 98.33% coverage with 18 SOOAs, 19.47 m/s cumulative Δv, and 9.91 m minimum clearance, reducing Δv by 9.85% relative to the dynamics-aware incumbent. A matched component ablation attributes this primary gain to local sequence search: local search alone returns the same route in 52.81 s, critic-only planning requires 26 actions and 42.86 m/s, rollout gives 20.92 m/s, and the complete planner takes 339.92 s. The safeguard prevents the critic from degrading a feasible base tour, while a changed initial state shows that the learned proposal can supply a safe fallback when the incumbent fails passive drift. Exhaustive Bellman recursion gives zero cost gap on nine tested 8–12-node reduced graphs. The evidence therefore supports safety-shielded graph policy improvement, not universal superiority of the learned critic. ROS 2 and Gazebo provide replay and interface evidence; quantitative claims remain tied to archived HCW logs.**

***Index Terms*— spacecraft inspection, approximate dynamic programming, policy iteration, safe observable orbital arc, relative orbital dynamics, passive safety, coverage planning, Hill–Clohessy–Wiltshire equations**

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I. Introduction

Large orbital structures are increasingly important to crewed flight, science, logistics, and in-space assembly. Their external surfaces must be inspected for configuration changes, impact damage, degraded thermal protection, docking-interface anomalies, and other faults that fixed onboard cameras may not observe. An autonomous inspector must therefore decide both where to look and how to move. These decisions are tightly coupled in orbit because a camera pose with high geometric coverage can be unreachable, control-intensive, or unsafe under relative motion.

Existing planning abstractions often expose only one side of this coupling. Robotic next-best-view methods provide mature treatments of sensor range, occlusion, incidence angle, and marginal coverage, but their motion cost is frequently Euclidean or platform-specific. Relative-orbit design provides dynamically meaningful trajectories, yet surface coverage and camera dwell are often secondary constraints rather than the planning objective. A sequential pipeline that selects views first and checks transfers afterward can therefore overestimate useful coverage. The selected views may be attractive in isolation while the connecting trajectory violates input or keep-out limits.

The deeper issue is the definition of a selectable inspection action. A static viewpoint is incomplete because it contains no evidence that the spacecraft can reach and dwell at that pose. A dynamically natural orbit segment is also incomplete when it does not reveal new weighted surface area. For orbital inspection, the action itself should include the transfer, the stabilized camera observation, and the safety evidence used to admit that action. This requirement motivates a finite representation that binds observability and relative-motion feasibility before the coverage sequence is optimized.

We introduce the safe observable orbital arc (SOOA), a finite-duration HCW transfer-to-dwell action annotated with its camera-valid target set, control effort, minimum clearance, terminal error, input and speed status, and optional passive-drift audit. Geometric viewpoints remain sampling seeds, but they do not contribute coverage until a corresponding transfer has passed the enabled feasibility tests. Figure 1 summarizes the resulting progression from mesh targets and camera visibility to a directed graph of admissible inspection actions. The graph objective then trades newly covered weighted area against the cost and margin of each reachable arc.

The finite representation makes sequential optimization and verification explicit. Candidate nodes are decision nodes, but a node identifier alone is not Markov because the value of a view depends on what has already been covered and how much budget remains. We therefore augment the graph state with covered- and selected-set masks and apply safety-shielded candidate-graph policy improvement. A learned critic, finite lookahead, rollout, and local search generate alternatives, while an audited base policy supplies a feasible completion and a no-degradation safeguard. Exact dynamic programming remains available only for reduced graphs small enough to exhaust.

The contributions are:

1) a finite-duration SOOA representation that unifies HCW reachability, camera visibility, dwell coverage, control effort, clearance, terminal accuracy, and passive-drift audit fields;
2) a Markov candidate-node formulation and safety-shielded policy-improvement algorithm combining a learned linear critic, truncated Bellman lookahead, incumbent rollout, and audited local search;
3) proofs of sampled compositional safety, incumbent no-degradation, a finite-horizon approximate-action-value bound, and exact optimality on exhausted reduced graphs;
4) matched component, primary, exact-oracle, initial-condition, and compute studies that isolate critic, rollout, safeguard, and local-search contributions and expose their planning-time costs; and
5) a reproducible separation between simulation and visualization in which the experiment writes archived data once and each paper figure is regenerated by a dedicated plotting function.

II. Related Work

Relative-motion models provide the dynamic foundation for orbital inspection. The Clohessy–Wiltshire equations remain a common baseline near a circular chief orbit because they are linear, interpretable, and inexpensive to propagate [1], [2]. Higher-fidelity flight design must account for nonlinear gravity, perturbations, navigation error, and actuator dynamics, but HCW propagation is useful for isolating how view ordering changes control effort and safety under a fixed model. Visibility-constrained orbit design has also been studied directly. Hou et al. analyze relative orbits for visual inspection of the China Space Station [3], while Zhou et al. combine near-natural relative ellipses with formation control around a rotating station model [4]. These approaches provide structured orbital motion and stronger closed-loop control analysis than the present planner, but their primary object is an orbit or formation rather than a sequence of surface-coverage actions.

Robotic inspection provides the complementary view-selection perspective. Classical next-best-view work models sensor range, incidence, occlusion, reconstruction quality, and motion cost [5]. Sampling-based kinodynamic planners such as RRT and asymptotically optimal variants incorporate differential constraints and collision avoidance in continuous state space [6], [7]. These methods are valuable when online reachability or continuous obstacle avoidance dominates. However, surface inspection additionally requires persistent accounting of which weighted targets have satisfied visibility and dwell. That reward must be attached to the motion primitive or repeatedly recomputed during search.

Approximate dynamic programming addresses sequential decisions when exact Bellman recursion is too large. Rollout evaluates a candidate action by completing it with a base policy and can inherit a policy-improvement guarantee in deterministic finite-horizon problems [8], [9]. Learning-based policies nevertheless require a separate safety argument: constrained policy optimization controls expected constraint costs [10], whereas shielding removes actions that violate an independently specified safety model [11]. OrbInspect uses the latter separation. Learning ranks only candidate actions that have passed a model-based HCW, input, clearance, terminal, and passive-drift audit; it is not trusted to infer safety from a scalar penalty.

Recent spacecraft-inspection studies increasingly couple geometry and motion. Apodaca et al. address inspection trajectory generation and coverage–energy trade-offs for in-orbit structures [12]. Zhou et al. study surface-aware tomographic paths around complex spacecraft geometry [13]. Ogri et al. address an execution-level problem involving image-feature tracking, information gain, and dwell under switched-system logic [14]. Together, these studies show that geometric coverage, orbital dynamics, and online sensing cannot be treated as fully independent. They also differ in the level at which the coupling occurs: orbit design, continuous path generation, online camera control, or discrete inspection planning.

OrbInspect is positioned at the discrete-action level. Its contribution is neither a new relative dynamics model nor a flight-certified collision-avoidance controller. Each SOOA is simultaneously a sensor-valid coverage contributor and a dynamically audited finite motion, so an infeasible transfer cannot improve the planner's reported coverage. This representation gives more detailed camera and sequential view-selection logic than formation-level observation studies, while those studies provide lessons for future work: near-natural relative arcs can seed lower-effort SOOA libraries, and a rotating body frame should replace the fixed-LVLH target assumption when station attitude motion is material. The current optimality claim remains conditional on the sampled library, and continuous-space global optimality is outside scope.

III. System and Observation Model

The mission is divided into offline planning and execution replay. Offline planning receives a known station mesh, a safe initial relative state, camera parameters, a keep-out model, and actuation limits. It returns a time-indexed translational reference, a camera-attitude stream, selected SOOA records, and summary metrics. Replay publishes these saved quantities without changing the planned state. This separation prevents visualization software from becoming an unreported dynamics backend.

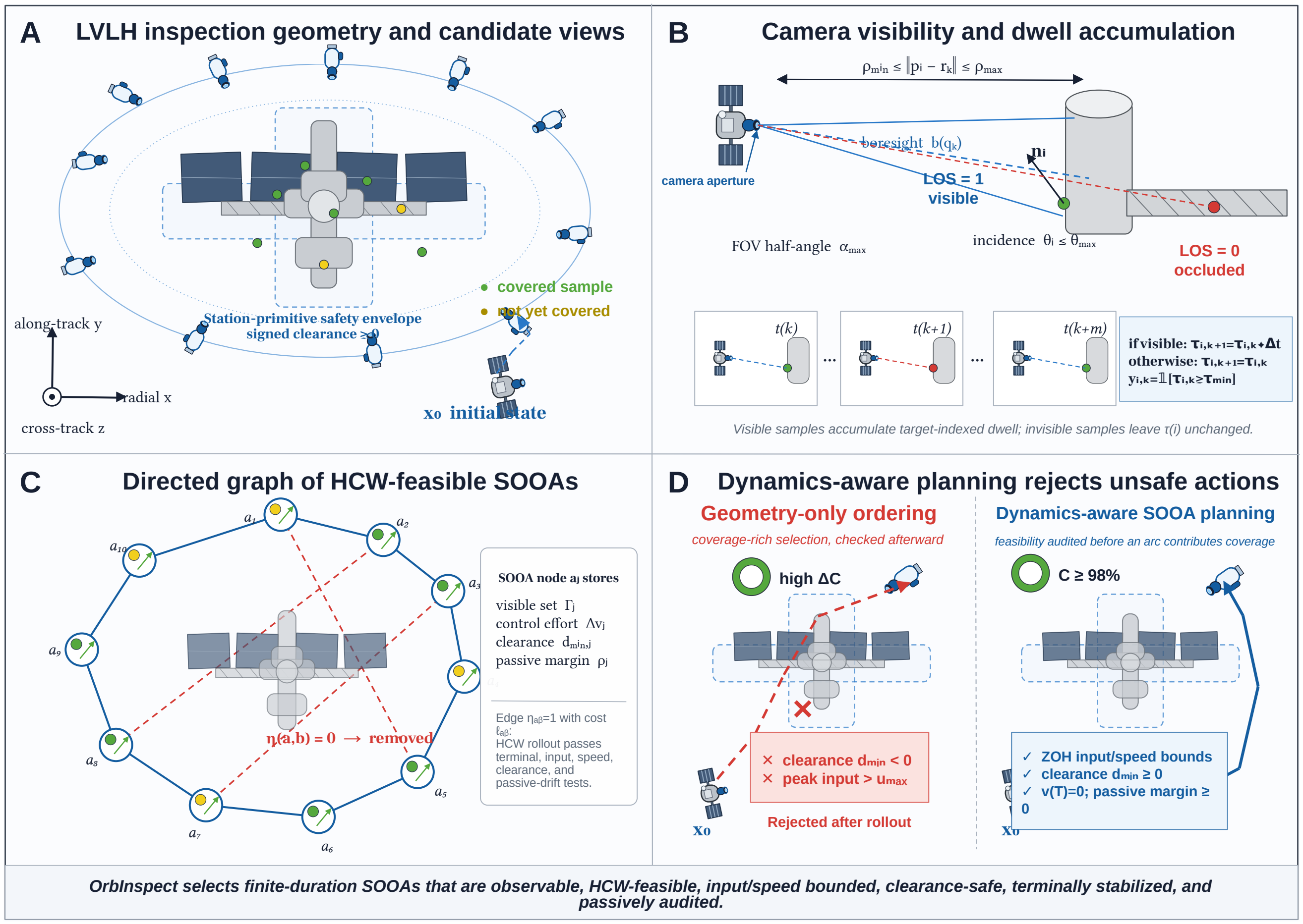

Fig. 1. OrbInspect formulation. (a) Candidate views are generated around an ISS-scale structure and outside a station-primitive safety envelope. (b) Range, field of view, incidence, line of sight, and dwell determine visibility. (c) Accepted SOOAs form a directed graph with HCW-feasible edges and coverage rewards. (d) Dynamics-aware selection rejects coverage-rich actions that violate input or clearance limits.

A. Relative Dynamics

The inspected structure is fixed in an LVLH frame attached to a nominal circular chief orbit. The axes follow the radial, along-track, and cross-track directions shown in Fig. 1. The chaser state is $\mathbf{x} = [\mathbf{r}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$, where $\mathbf{r} = [r_x, r_y, r_z]^\top$ and $\mathbf{v} = [v_x, v_y, v_z]^\top$. With mean motion n and commanded acceleration $\mathbf{u} = [a_x, a_y, a_z]^\top$, the HCW model is

$$
\begin{aligned}
&\dot{\mathbf{x}} = A_{\mathrm{HCW}}\mathbf{x} + B\mathbf{u},\\
&A_{\mathrm{HCW}} = \begin{bmatrix} 0 & I \\ K_n & C_n \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ I \end{bmatrix},\\
&K_n = \mathrm{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.
\end{aligned} \tag{1}
$$

Piecewise-constant acceleration over Δt gives the exact zero-order-hold propagation

$$
\begin{aligned}
\mathbf{x}_{k+1} &= A_d\mathbf{x}_k + B_d\mathbf{u}_k,\\
A_d &= e^{A_{\mathrm{HCW}}\Delta t},\\
B_d &= \int_0^{\Delta t} e^{A_{\mathrm{HCW}}\tau} B \, d\tau.
\end{aligned} \tag{2}
$$

Thus, the transfer model is finite-duration continuous maneuvering represented by sampled zero-order-hold inputs, not an impulsive pulse-mobility model. The command is continuous in mission time and constant only within each integration interval. Each segment is solved as a rest-to-rest transfer whose terminal position is the camera dwell pose and whose terminal relative velocity is zero. Independent segments are concatenated after their terminal audits, which explains the visible corners at some dwell transitions.

B. Visibility, Dwell, and Coverage

The structure is represented by weighted surface samples $\mathcal{T} = \{(p_i, n_i, w_i)\}_{i=1}^{N}$. Each weight approximates the surface area represented by its sample, so a densely tessellated component does not dominate solely because it contains more vertices. The camera attitude q_k defines unit boresight $b(q_k)$, and $\mathcal{O}$ denotes the occluding mesh.

TABLE I
Modeling assumptions and the boundary of the reported evidence.

Assumption	Role in the baseline	Consequence for interpretation
Circular chief orbit	Supports linear HCW relative translation in the LVLH frame.	Quantitative results compare planners under the same local linear model, not a perturbed flight ephemeris.
Station fixed in LVLH	Keeps mesh targets and safety primitives stationary during one planning run.	Rotating appendages or station attitude motion require a body-frame extension.
Known mesh and state	Makes visibility, occlusion, and transfer evaluation deterministic.	Navigation, map, and shape uncertainty are not represented by the present safety margin.
Prescribed camera attitude	Points the boresight toward each dwell target and exports the corresponding quaternion stream.	Camera slew time, reaction-wheel limits, and closed-loop attitude error are outside the optimization.
Sampled safety envelope	Computes signed clearance to station primitives at every rollout sample.	Positive sampled clearance is a planning audit rather than a continuous-time collision certificate.
No illumination or plume model	Isolates view geometry and translational feasibility.	Sun angle, shadow, contamination, and plume-impingement restrictions must be added for flight design.

Target i is visible when

$$\begin{aligned} \rho_{\min} \le \|p_i - \mathbf{r}_k\| &\le \rho_{\max}, \\ \arccos\frac{(p_i-\mathbf{r}_k)^{\mathsf{T}} b(q_k)}{\|p_i-\mathbf{r}_k\|} &\le \alpha_{\max}, \\ \arccos\frac{(\mathbf{r}_k-p_i)^{\mathsf{T}} n_i}{\|\mathbf{r}_k-p_i\|\,\|n_i\|} &\le \theta_{\max}, \\ \mathrm{LOS}(\mathbf{r}_k, p_i, \mathcal{O}) &= 1. \end{aligned} \tag{3}$$

The four tests exclude targets that are too near or too far, outside the field of view, observed at an excessive incidence angle, or blocked by station geometry. The dwell accumulator satisfies $\tau_{i,k+1} = \tau_{i,k} + \Delta t$ when visible and remains unchanged otherwise. A sample is covered when $y_{i,k} = \mathbf{1}\{\tau_{i,k} \ge \tau_{\min}\}$, and total weighted coverage is

$$C_k = \frac{\sum_i w_i y_{i,k}}{\sum_i w_i}. \tag{4}$$

Candidate generation can make a subset of the sampled surface unreachable by every accepted camera pose. We therefore also report inspectable coverage using $\mathcal{T}_{\text{insp}} = \cup_{j\in\mathcal{A}} \Gamma_j$:

$$C_k^{\text{insp}} = \frac{\sum_{i\in\mathcal{T}_{\text{insp}}} w_i y_{i,k}}{\sum_{i\in\mathcal{T}_{\text{insp}}} w_i}. \tag{5}$$

The total coverage in Eq. (4) answers how much of the sampled structure was inspected. Equation (5) answers how effectively the selected tour used the accepted camera library. Both are needed in density studies because coarsening the candidate set can change the denominator of the second metric. Attitude determines the visibility predicate and exported replay stream; closed-loop attitude dynamics and illumination are not optimized in this baseline.

C. Dynamic Feasibility and SOOAs

Every sampled state must satisfy

$$\|\mathbf{u}_k\|_2 \le u_{\max}, \qquad \|\mathbf{v}_k\|_2 \le v_{\max}, \qquad d_{\mathcal{S}}(\mathbf{r}_k) \ge d_{\text{safe}}, \tag{6}$$

where $d_{\mathcal{S}}$ is signed clearance from the station safety envelope. A transition rollout π_{ab} is accepted only if all samples satisfy Eq. (6) and its terminal position and velocity errors meet tolerance. The binary indicator $\eta_{ab} = 1$ records acceptance; otherwise the directed edge is removed or assigned a prohibitive cost for a deliberately unsafe baseline. The accepted edge cost combines control effort, duration, clearance penalty, and terminal error:

$$\begin{aligned} \ell_{ab} = \lambda_v \sum_{k\in\pi_{ab}} \|\mathbf{u}_k\|_2\,\Delta t + \lambda_t T_{ab} \\ + \lambda_s \sum_{k\in\pi_{ab}} \phi(d_{\mathcal{S}}(\mathbf{r}_k)) + \lambda_e e_{ab}. \end{aligned} \tag{7}$$

A SOOA over $[0, T_j]$ is

$$a_j = \left(\mathbf{x}_j(t), \mathbf{u}_j(t), q_j(t), \Gamma_j, \Delta v_j, d_j^{\min}, \rho_j\right), \tag{8}$$

where $\Gamma_j \subseteq \mathcal{T}$ is its dwell-valid target set, $\Delta v_j = \int_0^{T_j} \|\mathbf{u}_j(t)\|_2\,dt$, $d_j^{\min} = \min_t d_{\mathcal{S}}(\mathbf{r}_j(t))$, and ρ_j stores an optional passive-drift audit. The tuple is an action record rather than only a state: it contains the motion used to reach the observation and the evidence used to accept it. The audit propagates $\dot{\mathbf{x}}_{\text{drift}} = A_{\text{HCW}}\mathbf{x}_{\text{drift}}$ from sampled states over a prescribed horizon and rejects negative margin when enabled. It remains a sampled planning audit, not a flight-safety certificate.

IV. Safety-Shielded Graph Policy Iteration

A. Candidate Graph and Markov State

Let $\mathcal{A} = \{a_j\}_{j=1}^M$ be the SOOA nodes, let $G_j \subseteq \{1, \ldots, N\}$ be the target set visible during the dwell of node j, and let e_{ij} be the audited directed transfer from node i to j. The initial state is represented by node 0. Each edge stores stage cost ℓ_{ij}, minimum clearance, requested peak input, terminal error, and passive-drift margin. An edge is absent when any enabled audit fails.

A node identifier is not a sufficient decision state because the value of a view changes after its targets have been covered. The finite-horizon Markov state is

$$s_k = (j_k, m_k, b_k, h_k), \tag{9}$$

where j_k is the current node, $m_k \in \{0,1\}^N$ is the covered-target mask, $b_k \in \{0,1\}^M$ is the selected-node mask, and

h_k is the remaining action budget. Selecting node a gives the deterministic transition

$$f(s_k, a) = (a,\ m_k \vee G_a,\ b_k \vee \mathbf{e}_a,\ h_k - 1)\,. \tag{10}$$

The weighted coverage function is $C(m) = \sum_i w_i m_i / \sum_i w_i$, and $\mathcal{U}_s(s)$ denotes the shield-admissible action set. A state is terminal with zero cost when $C(m) \geq C_{\text{req}}$ and receives terminal penalty $\Psi(s) = \lambda_C[0.5 + (C_{\text{req}} - C(m))/C_{\text{req}}]$ if the horizon or safe action set is exhausted before the goal. For the undiscounted deterministic problem, the Bellman recursion is

$$V_h^\star(s) = \begin{cases} 0, & C(m) \geq C_{\text{req}}, \\ \Psi(s), & h = 0 \text{ or } \mathcal{U}_s(s) = \varnothing, \\ \min\limits_{a \in \mathcal{U}_s(s)} Q_h^\star(s,a), & \text{otherwise}, \end{cases}$$
$$Q_h^\star(s,a) = \ell(s,a) + V_{h-1}^\star(f(s,a)). \tag{11}$$

B. SOOA Construction and Safety Shield

Area-weighted targets are sampled from the mesh, and camera seeds are generated on offset shells around their outward normals. The binary visibility matrix $V_{ij} = \mathbf{1}\{i \in G_j\}$ is computed once after range, field-of-view, incidence, occlusion, and dwell tests. For a transition to dwell position $\bar{\mathbf{r}}_j$, the planner solves the minimum-energy piecewise-constant HCW boundary-value problem

$$\begin{aligned} \min_{\mathbf{u}_0,\ldots,\mathbf{u}_{K-1}} \quad & \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2^2 \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \\ & \mathbf{x}_K = [\bar{\mathbf{r}}_j^\mathsf{T}, \mathbf{0}^\mathsf{T}]^\mathsf{T}. \end{aligned} \tag{12}$$

Requested input is evaluated before clipping, and the resulting trajectory is propagated sample by sample. The shield admits

$$\mathcal{U}_s(s) = \left\{ a : \begin{array}{c} b_a = 0,\ G_a \setminus m \neq \varnothing,\ \eta_{ja} = 1, \\ d_{ja}^{\min} \geq 0,\ u_{ja}^{\max} \leq u_{\max},\ \rho_{ja} \geq 0 \end{array} \right\}, \tag{13}$$

where η_{ja} also contains the speed and terminal-state checks. The passive condition is omitted only when that audit is disabled. Unlike a reward penalty, the shield makes an inadmissible action unavailable to both training and execution [11].

C. Critic, Lookahead, and Safeguarded Improvement

Exact recursion is exponential in the mask dimensions. We approximate the action value by $\widehat{Q}_w(s,a) = w^\mathsf{T}\phi(s,a)$, where the ten normalized features contain a bias, post-action coverage gap and its square, stage cost, marginal weighted coverage, coverage-per-cost efficiency, used-budget fraction, inverse remaining horizon, clearance deficit, and input ratio. Safe exploratory rollouts generate Monte Carlo returns $R_k = \ell_k + R_{k+1}$ with the terminal penalty in Eq. (11); normalized stochastic-gradient updates fit w. The online selector expands a bounded branch of safe actions for d Bellman steps and uses $\widehat{Q}_w$ at the truncated leaves. This is approximation in value space rather than end-to-end policy learning [9].

Algorithm 1 Safety-shielded candidate-graph policy improvement

Require: SOOA nodes, lazy HCW edge evaluator, goal C_{req}, budget H
Build the deterministic feasible base policy μ_0
Fit $\widehat{Q}_w$ from shielded Monte Carlo rollouts
Generate π_{C} by depth-d critic-guided Bellman lookahead
for each rollout decision state s **do**
  Form shielded learned actions and all remaining base actions
  Evaluate Eq. (14) using audited base completions
  Select the minimum-cost action with a feasible completion
end for
Audit bounded reversal moves of the base sequence
Return the least-cost successful candidate among π_{C}, rollout, base, and improved base
Export policy source, critic errors, rejected actions, exact gap when available, and all trajectory records

The learned critic is not treated as a feasibility certificate. A deterministic two-stage coverage-and-HCW tour supplies a base policy μ_0. One-step rollout evaluates an admissible action by

$$Q_{\mu_0}(s,a) = \ell(s,a) + J_{\mu_0}(f(s,a))\,, \tag{14}$$

where J_{μ_0} is obtained by executing the remaining base sequence through the same shield. The base action is always included, so a feasible completion remains available. Audited reversal moves then search the incumbent sequence, and the returned policy is the least-cost coverage-satisfying member of the learned, rollout, base, and locally improved candidates. This safeguard is essential when critic approximation lowers partial cost without reaching the coverage goal.

D. Properties and Exact Oracle

PROPOSITION 1 (Markov sufficiency) *For a fixed deterministic SOOA graph, target weights, edge records, and horizon, the state in Eq. (9) is sufficient for the cost and feasibility of every future decision.*

Proof:
The current node determines the source of the next directed edge, m_k determines marginal and terminal coverage, b_k determines which nodes remain selectable, and h_k determines the remaining horizon. The transition, stage cost, shield, and terminal cost depend on no earlier quantity once these four fields are given. ■

PROPOSITION 2 (Sampled compositional safety) *Assume the initial state passes the enabled audits and every* *executed edge belongs to $\mathcal{U}_s$. Then the concatenated trajectory satisfies the input, speed, terminal, sampled-clearance, and enabled passive-drift conditions stored by the shield.*

Proof:
Each edge is accepted only after every stored sample and terminal state passes the enabled tests. Rest-to-rest terminal matching makes the terminal state of one accepted edge the initial state of the next. Induction over the selected sequence therefore preserves all stored sampled conditions. This proposition is not a continuous-time collision certificate between samples. ■

THEOREM 1 (Incumbent no-degradation) *Let μ_0 be a coverage-satisfying shield-feasible base sequence. If the returned candidate set contains μ_0 and every alternative is evaluated with the same stage costs, coverage goal, and shield, Algorithm 1 returns a feasible policy $\widehat{\pi}$ with $J(\widehat{\pi}) \leq J(\mu_0)$.*

Proof:
The base sequence is a feasible member of the final finite candidate set. Every other retained candidate has also reached the coverage goal through shielded edges. Selecting the least-cost retained candidate cannot have cost greater than the base member. ■

THEOREM 2 (Finite-horizon critic bound) *Let $\widehat{Q}_h$ approximate the optimal shielded action value at every reachable state with $|\widehat{Q}_h(s,a) - Q_h^\star(s,a)| \leq \epsilon$, and let $\pi_{\widehat{Q}}$ be greedy with respect to $\widehat{Q}_h$. For horizon H, $J(\pi_{\widehat{Q}}) - J(\pi^\star) \leq 2H\epsilon$.*

Proof:
At each state, approximate greediness makes the selected optimal action value at most 2ϵ above the minimum exact action value. Applying this inequality recursively to the deterministic successor state and summing over at most H decisions gives the bound. The result requires a uniform error bound; the experiments report empirical critic errors and do not claim that this assumption holds on the full graph. ■

For reduced-graph certification, exact recursion caches Eq. (11) for every reachable tuple (j, m, b, h). Bellman's principle then yields the following bounded claim.

THEOREM 3 (Optimality on an exhausted reduced SOOA graph) *For fixed finite nodes, visibility masks, audited edge costs, coverage requirement, and horizon, exhaustive evaluation of Eq. (11) returns a minimum-cost shield-feasible sequence on that graph.*

Proof:
The reachable state-action set is finite, Eq. (9) is Markov, and the recursion considers every admissible continuation. Backward minimization therefore returns the least-cost goal-reaching sequence. The claim does not extend to unsampled camera poses or continuous-space trajectories. ■

V. Experimental Protocol

A. Base Benchmark

The full benchmark uses the public NASA ISS glTF mesh [15]. The mesh is transformed into the planner's LVLH convention and sampled into 180 area-weighted targets. Offset-shell generation produces 357 camera candidates at a nominal 34 m radius. The initial state is $[0, -35, 10, 0, 0, 0]^\mathsf{T}$ in SI units, the keep-out margin is 2 m, the view budget is 36, and the training seed is 7. Every rest-to-rest segment lasts 90 s with a 2 s propagation step and a 0.060 m/s^2 input limit. Passive drift is audited for 300 s from every stored edge sample.

A 70% threshold defines mission success, whereas a 98% stop target forces methods to continue collecting difficult late-stage coverage. All methods use the same target weights, candidate library, visibility matrix, HCW propagation, keep-out model, input and speed limits, view budget, and termination logic. Each method begins with an empty HCW transfer cache, so planning time is not reduced by transfers evaluated for an earlier method. The main graph contains a coverage-preserving pool of 96 nodes; the critic uses 80 training episodes, branch width 6, candidate-audit width 18, and lookahead depth 2.

B. Matched Component Ablation

The component family repeats the primary 180-target graph with seven cold-cache variants. We denote critic training and finite lookahead by C, one-step rollout by R, the unchanged incumbent safeguard by S, and audited reversal search by L. The variants are the standalone incumbent; C only; C+S; R+S; L+S; C+R+S; and the complete C+R+S+L method. A disabled component is neither evaluated nor allowed to enter final policy selection. Consequently, solution quality and planning time measure the component set named by each row rather than diagnostics extracted after running the complete algorithm.

C. Oracle, Robustness, and Compute Cases

The exact-oracle family uses 48 targets, a seven-action budget, and 8, 10, or 12 candidate nodes. Its 35% requirement is deliberately below the full mission goal so every reduced graph contains a coverage-satisfying policy. Seeds 3, 7, and 11 change critic exploration but not the deterministic graph or exact Bellman solution. This family tests implementation optimality on exhaustible instances rather than full-coverage scalability.

The robustness cases IC-0, IC-1, and IC-2 use $\mathbf{x}_0 = [0, -35, 10, 0, 0, 0]^\mathsf{T}$, $[18, -42, 12, 0, 0, 0]^\mathsf{T}$, and $[-20, -32, 8, 0, 0, 0]^\mathsf{T}$, respectively. They retain 180 targets and compare the complete graph policy with the same two-stage base policy on a 72-node pool. The compute family fixes 90 targets and a 48-node pool, then compares zero training with depth 1, 20 or 80 training episodes with depth 2, and 80 episodes with depth 3. These cases expose

TABLE II
Matched baseline definitions used in the full benchmark.

Method	Selection rule	Purpose in the comparison
Complete graph policy	Algorithm 1 with critic, lookahead, shield, rollout, incumbent safeguard, and local search.	Tests long-horizon node decisions and safeguarded policy improvement.
Two-stage incumbent	Construct a feasible weighted-coverage subset and order it by feasible HCW dynamic cost.	Supplies the base policy and isolates improvement beyond the existing solver.
Coverage greedy	Select the largest new weighted area with only a small geometric-distance tie breaker.	Shows the failures hidden when coverage dominates transfer feasibility.
Safe coverage	Select the largest new weighted area among feasible transfers.	Separates feasibility filtering from effort-aware ordering.
Fuel greedy	Strongly penalize transfer Δv among feasible candidates.	Tests whether local effort minimization alone preserves high coverage.

cost, coverage, and runtime tradeoffs; they are not used to select a result after observing the full benchmark.

D. Saved Results and Replay

Each simulation case writes configuration, trajectory, control, coverage, safety, planner, and mission-event records into an independent results directory. The study root additionally stores one row per case and method, including learned, rollout, base, returned, and exact graph costs. The plotting module is read-only with respect to these records and provides one public function per standalone figure. This arrangement prevents stylistic revisions from changing the numerical experiment and makes every plotted value traceable to a saved row.

ROS 2 replay publishes saved odometry, path, controls, attitude, and camera frusta. Gazebo Harmonic supplies station and chaser visual context but does not propagate the orbital state. The quantitative evaluation is therefore identical with or without replay.

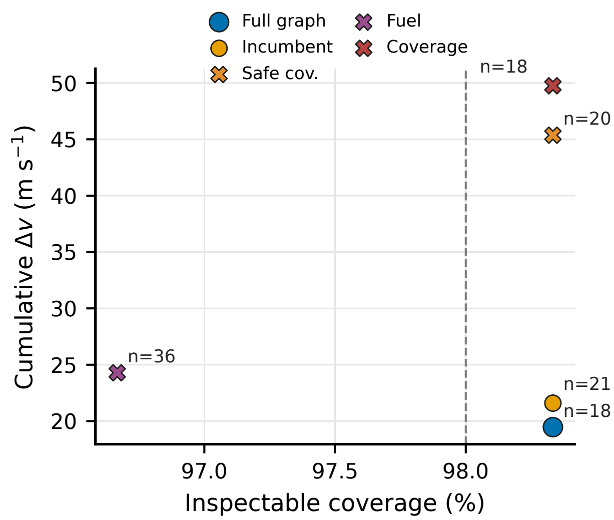

Fig. 2. Primary saved-result comparison. Position gives final inspectable coverage and cumulative Δv; annotations give selected action count, and the dashed line marks the 98% stop target. Circles pass every enabled trajectory and passive-drift audit, whereas X markers fail at least one audit.

VI. Results

A. Primary Coverage–Effort Comparison

The complete graph policy reaches 98.33% inspectable coverage with 18 SOOAs and 19.47 m/s cumulative Δv. The two-stage incumbent reaches the same coverage with 21 SOOAs and 21.59 m/s. The returned policy therefore removes three actions and reduces Δv by 9.85% while preserving 9.91 m sampled clearance and a positive 1.51 m passive-drift margin. Figure 2 places this result beside local baselines and shows action count directly rather than using marker area.

B. Component Attribution and Computational Cost

Table III changes the interpretation of the local baselines once passive drift is included in the common feasibility flag. Safe coverage reaches 98.33% without an instantaneous collision, yet its worst passive margin is -2.40 m; fuel greedy has the same failure and also misses the 98% stop target. Coverage greedy violates the input, instantaneous-clearance, and passive-drift checks. The complete graph policy is therefore compared with the two-stage incumbent as the only other primary method that both reaches 98% and passes every enabled audit.

Figure 3 reports matched cold-cache component runs rather than candidates extracted after executing the complete algorithm. Critic-only planning C nearly doubles incumbent Δv, from 21.59 m/s to 42.86 m/s, and increases the action count from 21 to 26. Adding the safeguard S restores the incumbent exactly but requires 278.99 s. Rollout with the safeguard R+S reduces Δv to 20.92 m/s with 20 actions, whereas local search with the safeguard L+S reaches 19.47 m/s with 18 actions. Adding critic and rollout to L does not change that route.

The full C+R+S+L method takes 339.92 s, 6.44 times the 52.81 s used by L+S, despite returning the same route and objective. C+R+S also reproduces R+S while increasing planning time from 97.64 s to 334.11 s. Thus local sequence improvement is the source of the primary-case gain, and the critic adds no measurable primary-

TABLE III
Primary 180-target benchmark. The feasibility column includes the enabled 300 s passive-drift audit.

Method	C (%)	SOOAs	Δv (m/s)	Peak u (m/s^2)	$d_{\min}$ (m)	$\rho_{\min}$ (m)	Plan (s)	Feas.
Complete graph policy	98.33	18	19.47	0.0385	9.91	1.51	339.92	Yes
Two-stage incumbent	98.33	21	21.59	0.0388	9.91	1.51	6.05	Yes
Coverage greedy	98.33	18	49.75	0.0849	-1.89	-3.43	5.35	No
Safe coverage	98.33	20	45.37	0.0581	0.89	-2.40	6.04	No
Fuel greedy	96.67	36	24.29	0.0368	0.14	-1.99	10.85	No

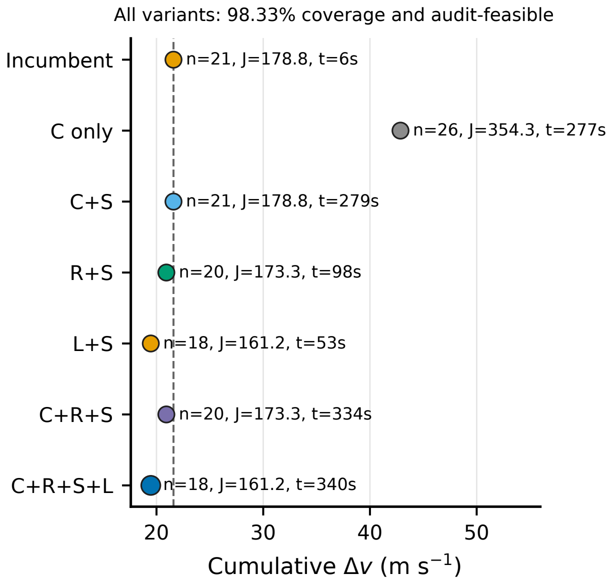

Fig. 3. Matched component ablation on the primary graph. C is critic training and lookahead, R is one-step rollout, S is the incumbent safeguard, and L is audited reversal search. Every variant reaches 98.33% coverage and passes the enabled audits. Position gives cumulative Δv; annotations report action count n, graph objective J, and cold-cache planning time t. The dashed line is the standalone incumbent.

case benefit. The complete method is an offline diagnostic architecture in its present implementation, not a real-time planner.

C. Reduced-Graph Certification

Across the tested 8-, 10-, and 12-node families, safeguarded rollout matches the exact Bellman cost for every seed, giving zero reported optimality gap. Figure 4 also reports the number of expanded Bellman states, which grows with graph size even under the modest seven-action horizon. Exploration seed changes the learned critic but not the deterministic graph optimum. These cases verify the state recursion and returned graph objective; they do not establish optimality on the 96-node primary graph.

D. Initial-State Robustness

Figure 5 separates effort from passive feasibility. At IC-0, policy improvement reduces Δv from 21.59 to 19.47 m/s; both policies retain a positive passive margin.

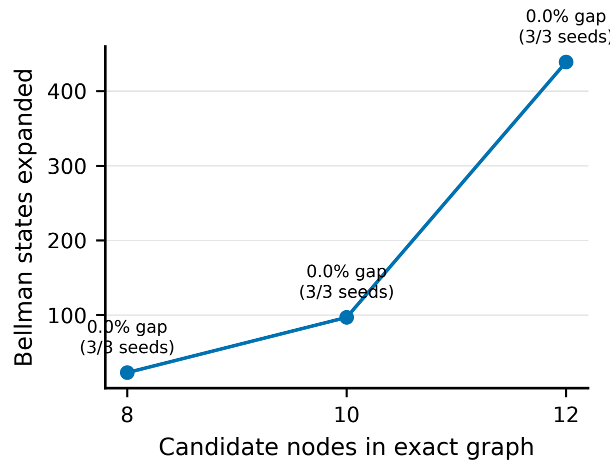

Fig. 4. Exact-oracle verification on reduced graphs. Vertical position is the number of Bellman states expanded; each point is annotated with the returned-policy cost gap to the exact optimum. Multiple seeds test the learned proposal while leaving the deterministic exact problem unchanged.

At IC-1, the two-stage incumbent is cheaper at 23.09 m/s but has $\rho_{\min} = -2.45\ m$. The shield rejects that completion, and the complete graph policy instead returns a 37.12 m/s route with $\rho_{\min} = 4.12\ m$. At IC-2, both policies have $\rho_{\min} = 6.86\ m$, and one-step rollout gives a small reduction from 20.19 to 20.10 m/s. The higher effort at IC-1 is therefore the cost of satisfying the enabled audit, not a performance improvement in fuel.

The three cases are deterministic stress tests rather than a statistical robustness guarantee. They show that the base policy can remain adequate, can be improved, or can become passive-unsafe as the start changes. The shield and safeguard respond to these cases differently: no-degradation applies only when the supplied base policy is itself coverage-satisfying and shield-feasible.

E. Critic and Lookahead Compute Tradeoff

Figure 6 shows that additional training is not monotonically beneficial on the truncated 48-node graph. With no Monte Carlo fitting and depth 1, the initial hand-scaled features reach 94.44% weighted surface coverage in 60.77 s. Twenty episodes with depth 2 cross the 95% target, reaching 95.56% in 83.98 s. Eighty episodes at the same depth reduce the partial-path objective but terminate at 92.22%, despite a smaller mean absolute training error.

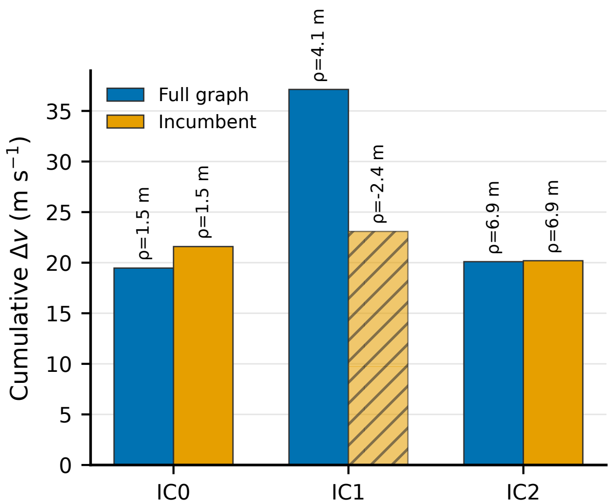

Fig. 5. Initial-condition comparison. Bars show cumulative Δv and annotations show minimum passive-drift margin $\rho_{\min}$; hatching denotes failure of an enabled audit. IC-0, IC-1, and IC-2 are defined explicitly in Section V.

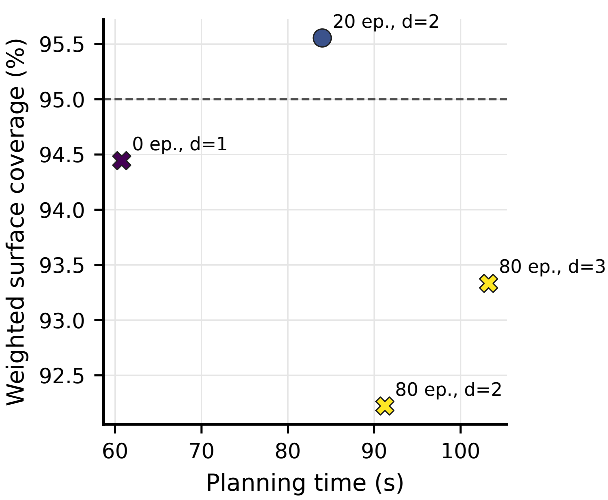

Fig. 6. Compute ablation on the 90-target, 48-node graph. Position gives cold-cache planning time and weighted surface coverage; annotations give training episodes and lookahead depth. Circles reach the 95% graph goal and X markers do not.

This behavior is consistent with value approximation error and distribution shift; it is why Algorithm 1 retains a success-aware incumbent rather than accepting the lowest predicted cost.

The depth-3 case tests whether more online search repairs the critic. Planning time rises from 91.23 s at depth 2 to 103.25 s, but coverage reaches only 93.33%; the additional expansion does not restore the target. These results rule out a blanket convergence claim for the linear critic and motivate future fitted-value or graph-neural approximators with validation on held-out initial states.

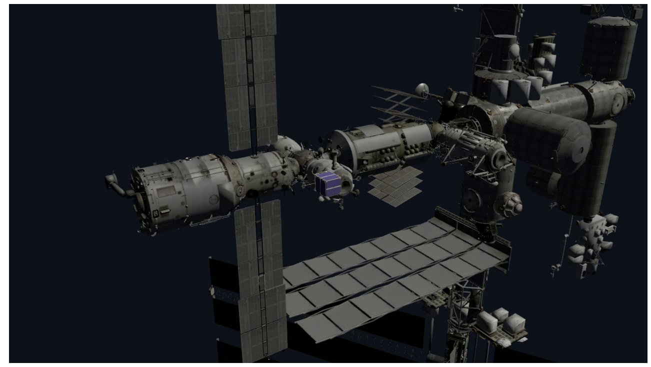

Fig. 7. Gazebo Harmonic replay context with the ISS model and the chaser shown near the central structure. The chaser pose is supplied by ROS from the saved planner trajectory; Gazebo dynamics are not used for quantitative evaluation.

F. Replay Consistency

Figure 7 shows the full station and chaser in Gazebo Harmonic. The chaser pose is driven by the ROS replay stream, while the station supplies visual context. The replay additionally publishes the saved camera boresight and frustum used by the offline visibility model. This verifies model loading, scale, frame publication, and pose-following interfaces; it does not independently validate image quality or orbital dynamics.

G. Cross-Result Synthesis

The results separate five questions that a final-coverage number would collapse. Figure 2 asks whether a successful shield-feasible policy reduces effort and action count. Figure 3 identifies which enabled component produces that gain. Figure 4 checks the finite-state implementation against exact Bellman recursion. Figure 5 tests whether a low-effort incumbent remains passively safe when the start changes, and Fig. 6 exposes the accuracy–runtime tradeoff of approximate values. Together they support a bounded conclusion: safeguarded local graph improvement can improve a feasible incumbent, and the broader architecture can replace an unsafe incumbent under the stated HCW and sampled-audit model; critic training alone is neither sufficient nor monotonically beneficial.

VII. Discussion

The candidate-node MDP strengthens the formulation beyond static next-best-view selection because covered area, current terminal node, and remaining budget enter the decision state explicitly. The reduced-oracle results verify that this state is sufficient for exact Bellman recursion. On the primary graph, however, the critic-only sequence is substantially worse than the incumbent, and the complete result is reproduced by safeguarded local search. The demonstrated primary advantage therefore comes from the safety-shielded graph representation and audited sequence improvement, not from an unsupported claim that reinforcement learning discovers a superior policy.

The passive audit changes which comparisons are meaningful. A transfer can satisfy input and instantaneous clearance while its uncontrolled continuation enters the protected region. Treating 'passive safe' as a separate diagnostic but excluding it from the common feasibility flag would make a low-effort unsafe tour appear superior. The corrected evaluation requires every method to pass the same enabled audit. IC-1 then shows the intended behavior: the complete graph policy accepts greater effort to replace a passive-unsafe incumbent. Because the primary ablation finds no critic benefit, this stress case supports only a conditional role for learned proposals when the base completion is inadmissible.

The primary cost reduction is empirical and library-conditional. Audited reversal improves the supplied base sequence, one-step rollout provides a smaller improvement, and the critic-only candidate degrades it. The 6.44-fold runtime gap between the complete method and L+S further argues against deploying the full stack without a case where the critic adds value. Different candidate densities, motion-primitive families, or critic features may change that attribution. The theorem guarantees no degradation only relative to a base sequence that is feasible on the same finite graph; it gives neither continuous-space optimality nor improvement when the base policy fails the shield.

The relation to formation-based observation is similarly complementary. Near-natural relative ellipses and closed-loop formation control around a rotating station can provide lower-effort motion structure and stronger stability analysis [4]. OrbInspect instead resolves which surface regions are visible, how dwell accumulates, and which observation should follow the current state. A useful synthesis would generate SOOA candidates from near-natural orbit families, then select among them using the present coverage and camera model. This could reduce transfer effort while preserving explicit target-level inspection accounting.

The safety evidence remains limited to the stated model. HCW translation assumes a circular chief and a fixed LVLH target. Clearance is checked on sampled trajectories against deterministic station primitives, and passive drift is audited only over a prescribed horizon. There is no closed-loop proof under navigation error, target rotation, delayed commands, actuator uncertainty, or between-sample collision. The exact oracle certifies only the exhausted reduced graphs, and the full solver has no continuous-space optimality guarantee.

The camera evidence has comparable boundaries. The visibility model includes range, field of view, incidence, occlusion, and cumulative dwell, but omits illumination, exposure, blur, plume contamination, and detailed attitude dynamics. Gazebo verifies model loading and replay interfaces, not orbital mechanics, collision avoidance, or image quality. A visually plausible replay is therefore not treated as quantitative validation.

A flight-oriented development path should first accelerate cached passive audits and batched edge evaluation, since the primary planning-time penalty is large. A held-out validation protocol and fitted or graph-structured critic should then replace the present linear Monte Carlo approximation. Uncertainty-inflated keep-out volumes, camera-slew and illumination constraints, plume restrictions, navigation error, and a rotating body frame should follow. Higher-fidelity propagation, including an optional Basilisk backend, can finally be introduced while retaining HCW as a transparent regression reference.

VIII. Conclusion

This paper formulated orbital inspection as safety-shielded sequential selection of finite-duration observable orbital arcs. The Markov state retains the current candidate node, covered targets, selected nodes, and remaining budget; critic-guided lookahead, rollout, and local search propose decisions, while an HCW shield and audited incumbent control feasibility and degradation. On the primary ISS graph, the returned policy reaches 98.33% coverage with 18 actions and 19.47 m/s cumulative Δv, a 9.85% reduction relative to the two-stage base tour at the same coverage and safety margins. A matched ablation shows that safeguarded local search reproduces this result in 52.81 s; the full method takes 339.92 s, and critic-only planning is worse than the incumbent. Reduced graphs match the exact Bellman optimum, and the IC-1 case shows why passive-drift feasibility must be evaluated alongside instantaneous clearance.

The evidence also defines the boundary of the contribution. The learned critic does not produce the primary gain, additional training is not monotonically beneficial on a truncated graph, and the full-stack planning-time overhead is substantial. Its role is presently conditional: it can supply an alternative when the incumbent is rejected, but its superiority is not demonstrated. Exact optimality is limited to exhausted reduced graphs; safety is limited to the deterministic sampled HCW, geometry, and passive-horizon audits. ROS and Gazebo verify replay consistency without replacing those planner dynamics. The result is therefore a transparent safety-shielded graph policy-improvement framework, not a flight-certified or continuously optimal solution.

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